Quest 1: The Coneservation

Lord of the Wings · topping up the pine-cone defences of Quackshire

> This guide uses the official puzzle text. The story is summarised and the rules are restated in plain words. It explains what the quest asks and the logic behind my solutions, not the code itself. Every number in the examples and traces was checked by running my solutions on the sample data (plus small instrumented copies to print intermediate values).

THE QUESTION
Each outpost has some pine cones. Find the smallest valid formation that uses all of them, and add up how many cones are missing.

INPUT
One comma-separated list of numbers: cones per outpost.

OUTPUT
The total number of new cones to collect.

WHAT CHANGES BETWEEN PARTS
Only the shape of a valid formation.

The story

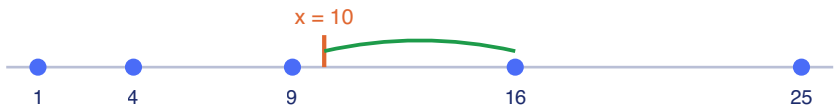
Before leaving Quackshire, the narrator and Frodoff check the village defences. The outposts protect the village with **formations of enchanted elven pine cones** laid out on a grid. Many cones have lost their magic, so each outpost has fewer than it needs. The outposts form **three defensive rings**, and each ring uses a different formation shape.

Understanding the question

Three rules apply in every part:

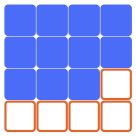
1. Each outpost builds **exactly one** formation.
2. It must use **all** of its cones. It can't leave any out or throw any away.
3. It can only **add** new cones.

So each outpost grows to the **smallest allowed size that is at least its cone count**, and collects the gap:

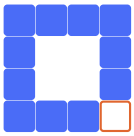


Part I sizes: 10 cones must jump to 16 → collect **6**

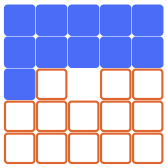
The three formation shapes



Ring 1: filled square
11 cones → 4×4, collect 5



Ring 2: square frame
11 cones → 4×4, collect 1



Ring 3: filled square, centre empty
11 cones → 5×5, collect 13

Blue = cones the outpost already has; orange outline = cones it must collect.

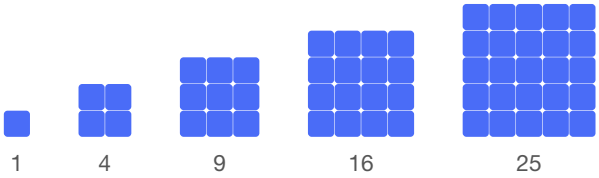
Ring	Valid sides n	Cones needed	Valid sizes
1 (Part I)	1, 2, 3, 4, …	n^2	1, 4, 9, 16, 25, 36, …
2 (Part II)	3, 4, 5, 6, …	$4(n - 1)$	8, 12, 16, 20, 24, 28, …
3 (Part III)	3, 5, 7, 9, …	$n^2 - 1$	8, 24, 48, 80, 120, …

Reading the input (same in all three parts)

The input is one list of numbers separated by commas (and possibly spaces or line breaks). Each number is one outpost's cone count, so `3,11,12` means three outposts with 3, 11 and 12 cones. All three parts share the same sample.

PART I Filled squares

A valid formation is a completely filled n×n square holding n^2 cones, any side $n \geq 1$.



Worked examples

Cones x	3	11	12	16	24	25	Total
Side n	2	4	4	4	5	5	
Holds	4	16	16	16	25	25	
Collect	1	5	4	0	1	0	**11**

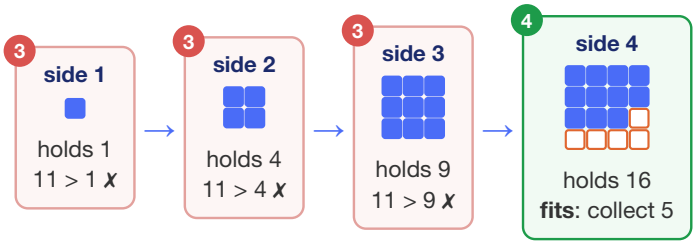
Step-by-step solution

1. Read the list of numbers `x` (one per outpost).
2. For one outpost, start with the smallest square: side `n = 1`.
3. If `n × n` is still smaller than `x`, the square is too small, so try `n + 1`. ↻ repeat
4. When `n × n ≥ x`, you have found the smallest square that fits. The cones to collect are `n × n − x`.

↻ *Do steps 2–4 for every outpost and add up the results. That total is the answer.*

How my solution works

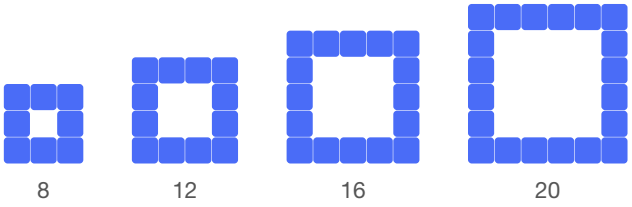
One outpost at a time: start at side 1 and grow the square until it holds all the cones. Here is an outpost with 11 cones:



Dots match the step numbers: ③ too small, grow; ④ it fits.

PART II Square frames

Only the border of a square, and it must surround at least one empty cell, so the smallest frame is 3×3.



Where 4(n – 1) comes from

Top row n + bottom row n + two sides of n – 2 each = **4(n – 1)**. A 5×5 frame: 5 + 5 + 3 + 3 = 16.

Worked examples

Cones x	3	11	12	16	24	25	Total
Side n	3	4	4	5	7	8	
Holds	8	12	12	16	24	28	
Collect	5	1	0	0	0	3	**9**

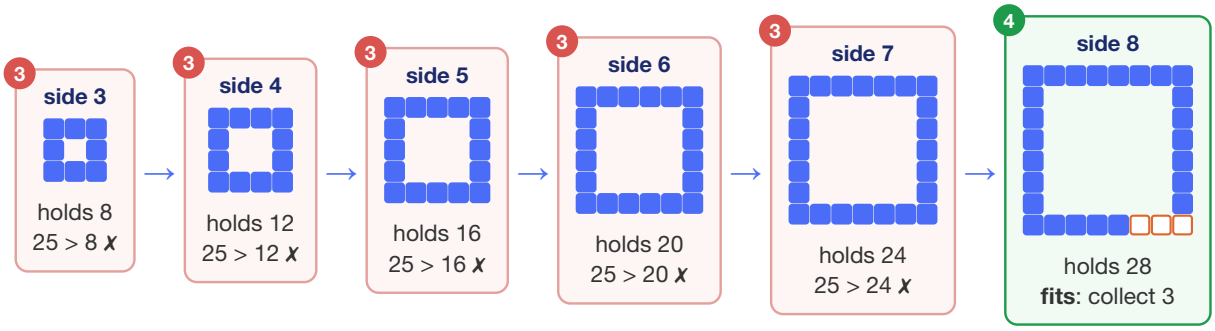
Frame sizes go up by 4 each time, so an outpost never needs more than 3 extra cones, except when it has fewer than 8.

Step-by-step solution

1. Read the numbers, same as Part I.
2. For one outpost, start with the smallest valid frame: `n = 3`. Smaller frames have no empty cell inside.
3. Work out the frame size `4 × (n − 1)`. If it is smaller than `x`, try `n + 1`. ↻ repeat
4. When `4 × (n − 1) ≥ x`, the cones to collect are `4 × (n − 1) − x`.
5. Add this up over all outposts.

How my solution works

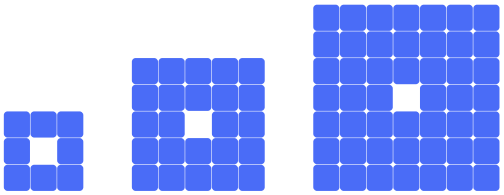
Same search as Part I, but it starts at side 3 and a side-n frame holds 4(n – 1). An outpost with 25 cones:



Dots match the step numbers: ③ too small, grow; ④ it fits.

PART III Filled squares with an empty centre

A filled square with exactly one empty cell in the very centre. Even sides have no single centre cell, so only odd sides work; a side-n formation holds $n^2 - 1$ cones.



8 24 48

Worked examples

Cones x	3	11	12	16	24	25	Total
Side n	3	5	5	5	5	7	
Holds	8	24	24	24	24	48	
Collect	5	13	12	8	0	23	**61**

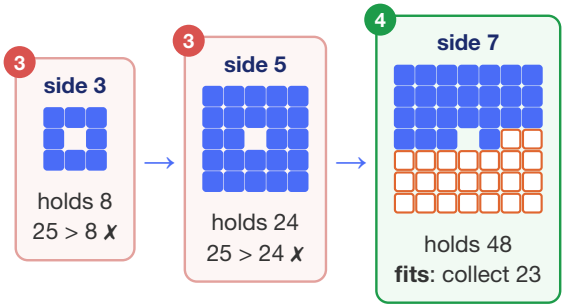
The gaps grow quickly because valid sizes jump by about 4n each step (8 → 24 → 48 → 80 …).

Step-by-step solution

1. Read the numbers, same as Part I.
2. For one outpost, start with `n = 3`, the smallest odd side.
3. Work out the formation size `n × n − 1`. If it is smaller than `x`, try `n + 2`, skipping even sides.
   ↻ repeat
4. When `n × n − 1 ≥ x`, the cones to collect are `n × n − 1 − x`.
5. Add this up over all outposts.

How my solution works

Same search again, but the side grows by **2** (3, 5, 7, …) so it stays odd. An outpost with 25 cones:



Dots match the step numbers: ③ too small, grow; ④ it fits.

Summary

Part	Start n	Step	Capacity	Sample
I	1	+1	n^2	11
II	3	+1	$4(n - 1)$	9
III	3	+2	$n^2 - 1$	61

All three parts use the same pattern: **start at the smallest valid side, grow until the formation is big enough, and return the gap**. The search only tries about $\sqrt{x}$ sides, which is instant for these inputs. You could also skip the search: in Part I the side is simply $\sqrt{x}$ rounded up.